

Neural Networks and Deep Learning Lab 9

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1 Introduction

In this lab, a Generative Adversarial Network (GAN) and a Conditional GAN (cGAN) were implemented using the Fashion-MNIST dataset to generate synthetic clothing images. The goal was to compare a standard GAN with a conditional GAN and analyze how conditioning on class labels affects the quality and controllability of generated images. Additionally, experiments were conducted by varying the latent space size, label embedding size, and label injection strategy to study their impact on image quality.

2 Standard GAN Results

**Standard GAN — Generated Images after 50 Epochs
(No class control — random class mix)**

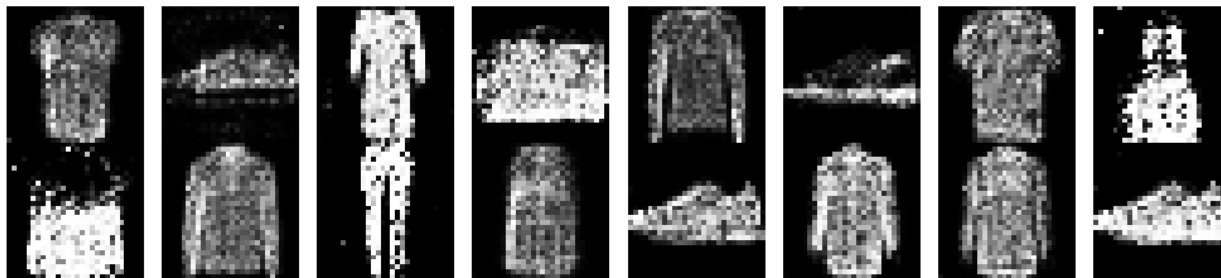


Figure 1: Images generated by Standard GAN

The standard GAN generates images without any control over class labels. The outputs consist of mixed clothing types and lack clear structure. Most images appear noisy and blurred, indicating that the model struggles to learn meaningful representations of the dataset. There is no way to enforce generation of a specific class.

3 Conditional GAN Results

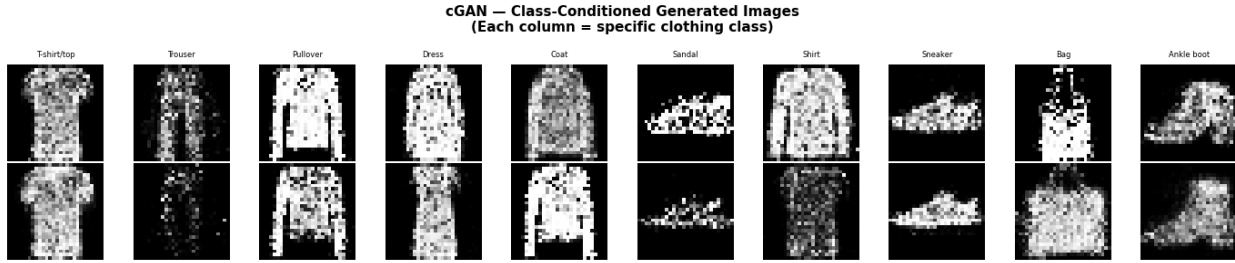


Figure 2: Images generated by Conditional GAN (each column corresponds to a class)

The conditional GAN allows generation of images corresponding to specific class labels. Each column represents a particular clothing category. Compared to the standard GAN, some class-dependent structures are visible (e.g., trousers appear vertical, shoes appear horizontal). However, the images are still noisy and not sharply defined, indicating that the model has not fully converged.

4 Comparison of GAN and cGAN

The standard GAN produces uncontrolled outputs, while the cGAN introduces class conditioning, enabling targeted generation. Although both models produce noisy images due to architectural limitations, the cGAN demonstrates better organization and partial class separation. Therefore, cGAN is better suited for controlled image generation.

5 Experiment 1: Latent Space Size

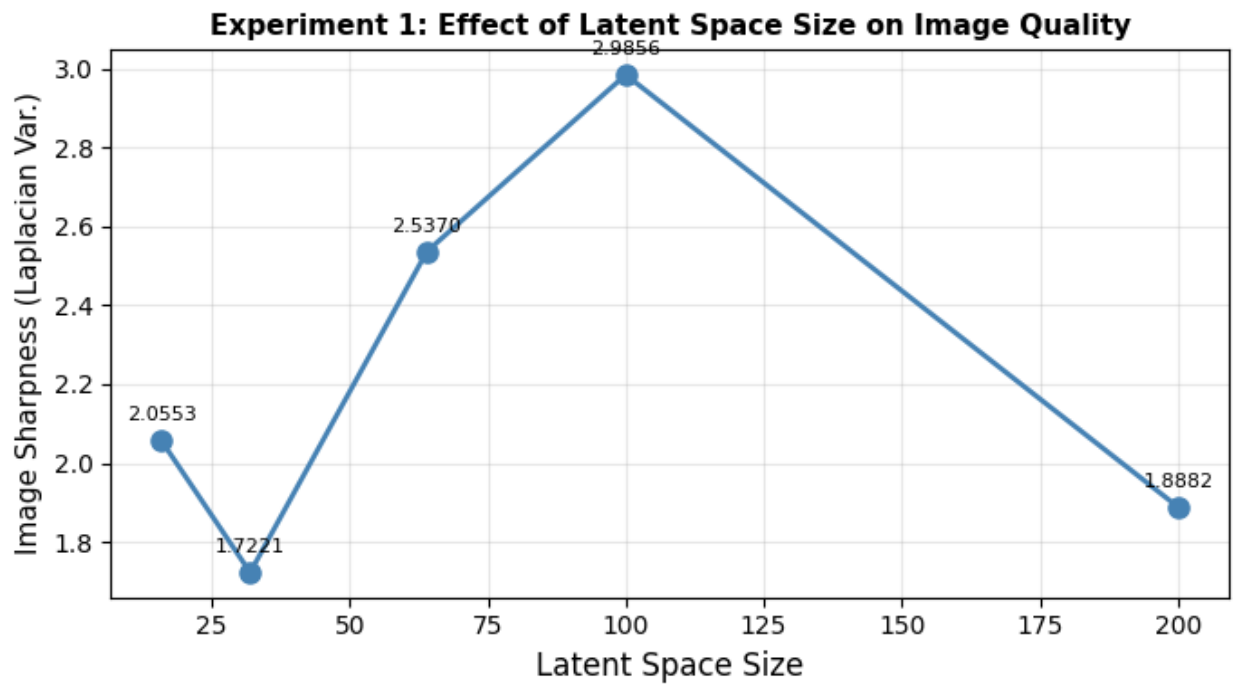


Figure 3: Effect of latent space size on image quality

As the latent dimension increases, image quality improves up to a certain point (around 100). Beyond this, the performance degrades due to increased complexity and instability in training. This indicates that there exists an optimal latent space size that balances diversity and learnability.

6 Experiment 2: Label Embedding Size

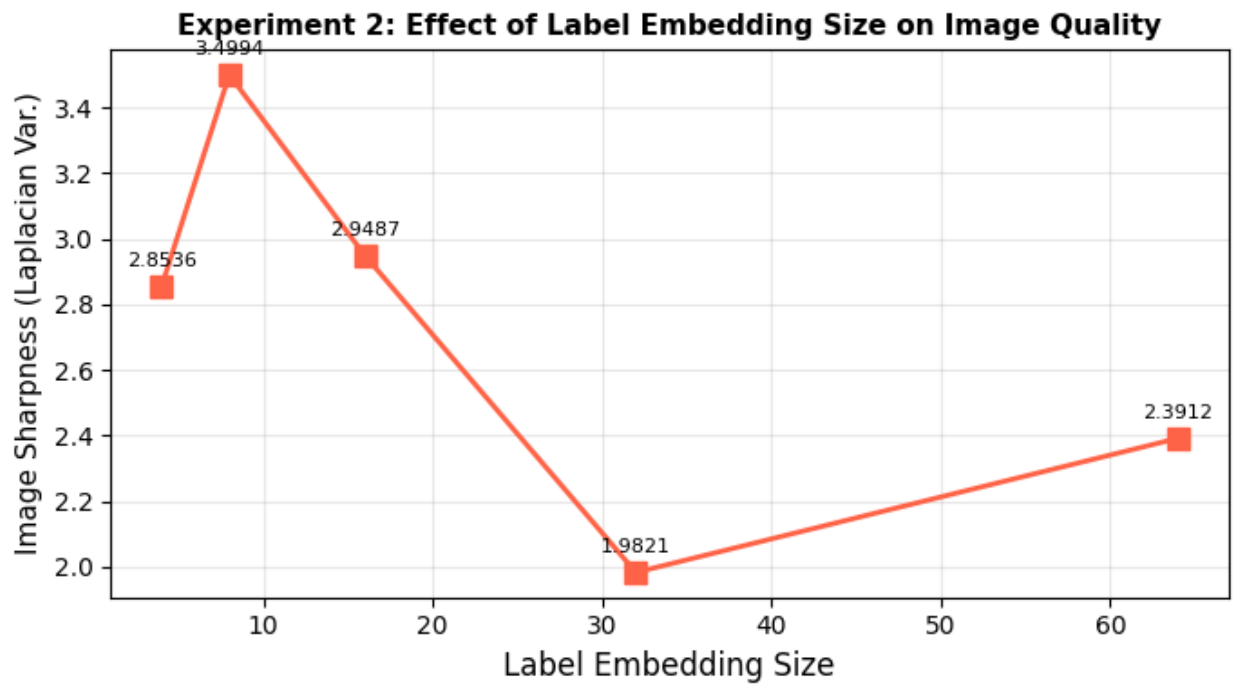


Figure 4: Effect of embedding size on image quality

The embedding size significantly affects the conditioning capability. Moderate embedding sizes perform better than very small or very large ones. Small embeddings fail to capture sufficient class information, while large embeddings introduce unnecessary complexity and noise.

7 Experiment 3: Label Injection Strategy

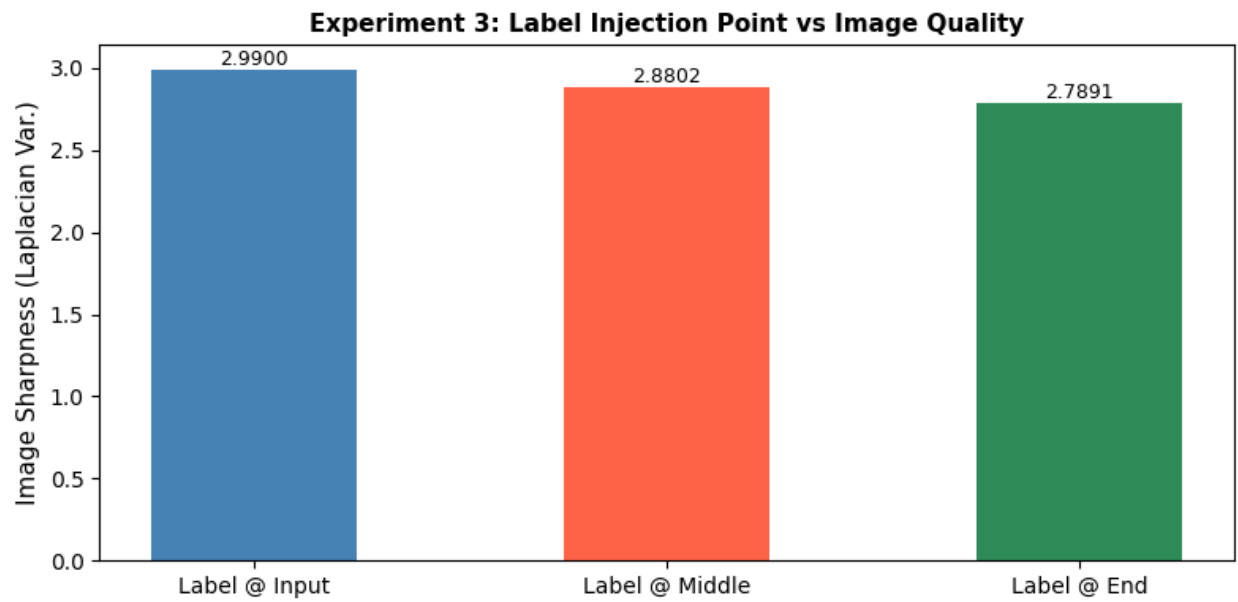


Figure 5: Effect of label injection point

Injecting label information at the input layer results in slightly better image quality compared to injecting at intermediate or final layers. This suggests that early conditioning helps guide the generator throughout the feature learning process.

8 Visual Comparison Across Experiments



Figure 6: Comparison of generated images across different experiments

From the visual comparison, it is observed that differences between experiments are not very prominent. Most generated images remain noisy and lack sharp details. This indicates that the limitations of the fully connected GAN architecture dominate the effect of hyperparameter changes.

9 Observations

- The standard GAN produces random, uncontrolled outputs with no class separation.
- The conditional GAN enables class-controlled generation, showing partial improvement in structure.
- Latent space size has an optimal range; too small reduces diversity, while too large affects stability.
- Label embedding size influences conditioning, with moderate values performing best.
- Injecting labels at the input layer is more effective than injecting at deeper layers.
- Overall image quality remains low due to the use of a simple fully connected architecture.
- Differences between experiments are not clearly visible because the model has not learned strong representations.

10 Conclusion

The conditional GAN is better suited for controlled image generation compared to the standard GAN, as it allows generation of images corresponding to specific class labels. However, both models produce noisy outputs due to architectural limitations. While hyperparameter tuning provides some improvements, it is not sufficient to achieve high-quality images. More advanced architectures such as convolutional GANs are required to generate realistic images.